

Project Design Phase – II

Solution Requirements (Functional & Non-Functional)

Date: 23 June 2026

Project Name: Metro Ticket Generating System in ServiceNow

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Metro Ticket Booking	Ticket booking through Service Catalog
		Source Station Selection
		Destination Station Selection
		Passenger Type Selection
		Number of Tickets Selection
FR-2	Fare Calculation	Automatic fare calculation based on source and destination
		Fare calculation based on passenger type
		Total fare generation
FR-3	QR Code Ticket Generation	Unique Ticket ID generation
		QR code creation for each ticket
		Digital ticket generation
FR-4	Ticket Confirmation	Confirmation message after booking
		Email notification with ticket details
		QR ticket delivery to passenger
FR-5	Station Management	Metro station master data management
		Source and destination validation
FR-6	Ticket History	View previously booked tickets
		Search ticket details
FR-7	Reporting & Analytics	Daily ticket sales report
		Revenue report generation
		Peak-hour passenger analysis
FR-8	Digital Payment Support	UPI payment integration
		Credit/Debit card payment support
		Transaction status tracking

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	User-friendly interface for easy metro ticket booking
NFR-2	Security	Secure storage of passenger and transaction data with role-based access control
NFR-3	Reliability	Accurate fare calculation and ticket generation with minimal system failures
NFR-4	Performance	Ticket booking and QR code generation should complete within a few seconds
NFR-5	Availability	System should be available 24x7 for passengers
NFR-6	Scalability	Support increasing passenger volume during peak travel periods
NFR-7	Maintainability	Easy updates to fare rules, stations, and ticketing configurations
NFR-8	Compatibility	Accessible from desktop, tablet, and mobile devices
NFR-9	Sustainability	Reduce paper usage through digital ticket generation
NFR-10	Data Analytics	Support reporting and dashboard generation for decision-making